

Use-Case Flow Specification

Municipal Infrastructure Damage Reporting App | UC-01: Report Infrastructure Damage

Primary Actor	Citizen Reporter
Supporting Actors	GIS Map Service; Municipal Engineer
Preconditions	The citizen can access the portal and has identified a pothole, broken streetlight, or water pipeline leak. The device has GPS access or the citizen can place a map pin.
Postconditions	A unique ticket is stored, routed to the matching municipal division, and a confirmation with the ticket ID is shown to the citizen.

Main Success Scenario

1. The citizen opens the portal and selects "Report Damage."
2. The system displays the report form and requests a damage category, description, photo, and location.
3. The citizen selects pothole, broken streetlight, or water pipeline leak; enters a short description; and attaches a photo.
4. The system extracts EXIF GPS data from the photo and displays the incident location on the municipal GIS map.
5. The citizen verifies the shown location and submits the report.
6. The system validates the required details, creates a unique ticket, and dispatches it to the matching municipal engineering division.
7. The system stores the report and displays a confirmation containing the ticket ID and initial status "Submitted."

Alternate Flow A1 - GPS metadata unavailable

- A1.1** At Step 4, if the uploaded photo contains no valid GPS data, the system informs the citizen that the location cannot be determined automatically.
- A1.2** The system displays the GIS map and asks the citizen to drop or adjust a manual location pin.
- A1.3** The citizen confirms the manual location; the use case resumes at Step 5.